Md Toufikul Islam Chowdhury

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EDUCATION

University of Greenwich
London, UK

2025 **M.Sc.**, Data Science, with Merit
January 2024 – February 2025 (awarded 24 February 2025)
Dissertation: 80% (distinction level)
Modules include Statistical Methods for Time Series Analysis; Machine Learning; Applied Machine Learning; Data Visualisation; Big Data

Leading University
Sylhet, Bangladesh

2022 **B.Sc. (Honours)**, Computer Science and Engineering
2018 – 2021 (degree conferred November 2022)
CGPA 3.30/4.00; 3.61/4.00 over the final two years
Modules include Probability and Statistics; Discrete Mathematics; Numerical Methods; Differential Equations and Fourier Analysis; Artificial Intelligence; Introduction to Sociology; Principles of Economics

MANUSCRIPTS IN PREPARATION

My research combines geospatial accessibility modelling, spatial statistics and statistical modelling to study social and geographic inequalities in access to maternity and neonatal care, using small-area population data and national administrative data for England. I carry out this work independently, in collaboration with Dr Jia Wang (University of Greenwich).

- Near but capacity-constrained: distance, capacity and urban pressure in access to neonatal intensive care in England. Md Toufikul Islam Chowdhury and Jia Wang. In preparation for *Urban Science* (MDPI).
  - 568,188 births in 33,609 LSOAs (July 2023 – June 2024); car travel times to 156 neonatal units on an OpenStreetMap road network re-weighted with observed 2024 Department for Transport speeds; intensive-care cot availability relative to births by the enhanced two-step floating catchment area method (E2SFCA).
  - Inequality measured across 18 population characteristics, including six deprivation domains, seven estimated ethnic groups of mothers and births to mothers aged 24 or under, using births-weighted concentration indices with decomposition and a Theil between/within decomposition.
  - Between- and within-catchment associations separated with a hybrid model and a conditional logit; cluster and wild cluster bootstrap inference, spatial lag and error models and false-discovery-rate control; trust nurse staffing modelled by quasi-binomial regression.
  - 17.5% of births were near a NICU but in areas of below-median modelled capacity; child income deprivation was associated with this state between catchments (OR 1.31 per SD, 95% CI 1.08–1.60) but not within them (0.99, 0.87–1.12).
  - Reproducible pipeline with 83 automated validation checks.

M.Sc. DISSERTATION

University of Greenwich
School of Computing and Mathematical Sciences

2024 – 2025

M.Sc. Dissertation Research
Mark: 80%
Supervisor: Dr Jia Wang

Spatiotemporal workforce dynamics and forecasting in NHS England: Spotlight on midwives and ambulance staff.

- Fifteen years of monthly NHS workforce statistics (2009–2024) for 42 integrated care systems; staff per 100,000 population mapped against the English Indices of Deprivation 2019, showing fewer midwives and ambulance staff in several deprived areas.
- Staffing forecast to 2029 with ARIMA, SARIMAX (with a COVID-19 exogenous term) and double exponential smoothing, compared by MAE, RMSE, MASE, AIC/BIC and rolling-origin two-year-ahead evaluation.
- Prototype interactive dashboard of workforce distribution and forecasts at regional and ICS level.

COMPETITIONS

- CUSP London Data Dive, King's College London, February 2025. Team 5 of 12 (over 80 students from eight universities), urban waste-management challenge. The team linked recycling, fly-tipping, deprivation (IMD), council tax and housing data for the 32 London boroughs and the City of London; I built the interactive borough map in Python. Recycling rates were lower in more deprived boroughs (Spearman $\rho = -0.63$) and in boroughs with more purpose-built flats ($\rho = -0.64$). Findings presented to a judging panel of public-sector and industry partners.

MISCELLANY

- Statistical methods: regression for binary and proportion outcomes (logit, conditional logit, quasi-binomial); cluster-robust and wild cluster bootstrap inference; time series (ARIMA, SARIMAX, exponential smoothing); inequality measurement (concentration index, Theil decomposition); multiple-testing control.
- Spatial methods: spatial autocorrelation (global and local Moran's I, join counts); spatial lag and error models; network travel-time modelling (OSRM); floating catchment accessibility (E2SFCA).
- Data: ONS (Census 2021, births, LSOA geography); NHS workforce statistics; National Neonatal Audit Programme; CQC; English Indices of Deprivation; Department for Transport; OpenStreetMap.
- Programming: Python (pandas, statsmodels, GeoPandas, PySAL, scikit-learn), SQL, QGIS, Git, LaTeX.
- Languages: English (fluent).

REFERENCES

- Dr Jia Wang, Senior Lecturer in Spatial Data Science, School of Computing and Mathematical Sciences, University of Greenwich. j.wang@greenwich.ac.uk
- Mohammad Jaber Hossain, PhD Candidate, Forestry and Renewable Energy Section, Norwegian University of Life Sciences (NMBU); formerly Lecturer, Leading University, Sylhet, Bangladesh. mohammad.jaber.hossain@nmbu.no